

Skill Arena - Phase 5 Document 5

Cloud Infrastructure, Deployment Architecture, CI/CD Pipeline, DevOps & Global Scaling Framework

Document Purpose

Define how the platform is deployed, operated, scaled and maintained in production environments.

Business Objective

Provide a reliable, scalable and continuously deployable platform capable of global operation.

Cloud Strategy

Cloud-native architecture designed for elasticity, resilience and automation.

Infrastructure Layers

Edge Layer, Application Layer, Service Layer, Data Layer and Operations Layer.

Deployment Architecture

Containerized workloads deployed through orchestrated infrastructure.

Environment Strategy

Development, Testing, Staging, Pre-Production and Production environments.

CI/CD Philosophy

Automated build, test, validation and deployment pipelines.

Source Control Strategy

Version-controlled repositories with branch protections and release workflows.

Build Pipeline

Code Validation, Security Scanning, Automated Testing, Artifact Creation and Release Packaging.

Testing Pipeline

Unit Tests, Integration Tests, Security Tests, Performance Tests and Regression Tests.

Deployment Pipeline

Automated deployment with approval workflows and rollback support.

Infrastructure as Code

Infrastructure definitions managed through version-controlled configuration.

Container Strategy

Standardized service packaging and deployment isolation.

Orchestration Platform

Automated scheduling, scaling, service discovery and workload management.

Auto Scaling Framework

Horizontal and vertical scaling based on workload demand.

Global Scaling Strategy

Multi-region deployments supporting worldwide users.

Load Balancing

Traffic distribution across services, regions and infrastructure zones.

Observability Platform

Centralized logging, monitoring, metrics, tracing and alerting.

Operational Dashboards

Infrastructure health, service health, deployment status and scaling metrics.

Release Management

Versioning, release tracking, deployment auditing and rollback controls.

Disaster Recovery Integration

Regional failover, backup restoration and operational recovery procedures.

Security Integration

Pipeline security scanning, secrets management and deployment validation.

Performance Monitoring

Track latency, throughput, resource consumption and user experience metrics.

Cost Management

Monitor infrastructure utilization, scaling costs and operational efficiency.

Compliance Requirements

Deployment audit trails, operational logging and governance controls.

Acceptance Criteria

Platform can be deployed, scaled, monitored and maintained reliably at global scale.